

1 DEVELOPMENT → TRAIN

Disjoint pool
no final labels

Validate contract

modes · templates · parser
retries · throughput · cost

Fixed cap

2,048 total
Report completion
Retain failures

Freeze D/R packages

manifest before every label

TRAIN

K draws/package
fit methods

2 TUNING

Disjoint tune set

fixed 1/6 family weights

Calibrate + search

same gate family, budget
resources, tie-to-direct

Select one

candidate per
learned method

Freeze candidate hash

no audit-set candidate change

3 HELD-OUT DEPLOYMENT AUDIT

Frozen candidate

one audit; no retuning

Empirical-Bernstein

score LCB · cost UCB
spoilage Z UCB
denominator B LCB

**All four
checks pass?**

Level-gamma intersection–union pass rule for one frozen candidate
Empirical safeguard—not conformal, pointwise, or shift guarantee

4 DEPLOYMENT / TEST

New input x

input only

Frozen gate

enabled only on pass
otherwise bypassed

Exactly one answer call

D: direct or R: bounded
native thinking

TEST: offline policy swap only

after action log; never changes deployment gate

PASS: enable frozen gate

FAIL: force D and bypass gate